

Carmen Lucca DiMario

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EDUCATION

Embry-Riddle Aeronautical University (ERAU)

Graduation:

Master of Science in Computer Science

GPA: 4.0

May 2027

Bachelor of Science in Computer Science

GPA: 3.97

May 2026

TECHNICAL SKILLS

Clearance: Held interim DoD Secret (inactive) | U.S. Citizen | FAA Part 107 Remote Pilot

Programming Languages: C++, C, Go, Python, Java, TypeScript, GNU ASM, Bash, SQL

Networking & Distributed: P2P networking, TCP/IP, REST APIs, Micro XRCE-DDS, UPnP/NAT traversal

Systems & Tools: Linux, UNIX, real-time embedded, STM32, ROS2, Docker, AWS, Git, TensorRT

PROFESSIONAL EXPERIENCE

Garmin: Embedded Software Development Intern

May 2026 - August 2026

- Implemented Clang compiler support replacing legacy ARMcc toolchains in embedded avionics software
- Extended embedded library support for 64-bit architectures, resolving data type alignment and pointer-width issues to broaden platform compatibility
- Developed a system test suite for a proprietary communication interface on a Garmin Integrated Avionics Unit (GIA), validating real-time data transmission and protocol correctness

JRC Integrated Systems: RISE Intern/CO-OP Participant

June 2025 - May 2026

- Developed and optimized software tools for radiation-hardened semiconductor testing
- Rebuilt the Accuro test suite from hard-coded shell scripts into a Python framework with parallel execution, selectable test subsets, and configurable output, cutting a multi-hour run to roughly 25 minutes
- Upgraded the Accuro application's PETSc integration across a major version boundary to enable GPU-accelerated solvers for its core computations

PROJECT EXPERIENCE

Dilithium: Post-Quantum Cryptocurrency (Go)

January 2026 - Present

- Built a peer-to-peer proof-of-work blockchain from first principles in Go, implementing the consensus, node networking, and block/transaction synchronization layers
- Ran competitive mining across a live P2P network with peer discovery and UPnP NAT traversal
- Signed every transaction with CRYSTALS-Dilithium Mode3 (NIST FIPS 204) and shipped cross-platform builds for Linux, macOS, and Windows

AIRHOUND: Autonomous UAV Target Tracking [Middleware Lead]

September 2024 - May 2026

- Designed a four-node ROS2 pipeline driving a PX4 flight controller over Micro XRCE-DDS
- Diagnosed DDS serialization as the pipeline bottleneck and replaced it with direct V4L2 capture, cutting mean detection latency from 234 ms to 58.8 ms and eliminating dropouts at 1280x720
- Deployed RF-DETR with TensorRT FP16 on a Jetson Orin at 15 FPS, sustaining the 10 Hz control loop
- Validated the full software stack in a 600-second SITL test at 30 Hz throughput across every pipeline stage

SEMBAS-RL: Graduate Research, Autonomous Driving Policy Assurance

June 2026 - Present

- Trained and evaluated thirteen RL driving policies (PPO and SAC) across five curriculum strategies over 5M+ training steps
- Extended SEMBAS, a three-phase adaptive boundary scanner, to RL policy evaluation over a 3D scenario space to characterize where a learned policy remains reliable
- Built three analysis agents that convert boundary geometry into ranked retraining proposals automatically

PUBLICATIONS

DiMario, C., Bacha, R., & Butka, B. "Combating Senior Scams Using a Large Language Model-Created Rubric." ASSE '24, ACM, 2024. <https://doi.org/10.1145/3702138.3702140>

Malarchick, R., DiMario, C., et al. "Predictive Target Pursuit for Autonomous UAVs Using RF-DETR with Depth-Aware State Estimation and Physics-Informed Trajectory Prediction." Proc. SPIE 14030, 140300O (2026). <https://doi.org/10.1117/12.3094732>

EXTRACURRICULAR ACTIVITIES

- **ERAU Cyber Rodeo 2026:** First place individual
- **ERAU Division II Cross-Country/Track:** Approximately 30 hours of training weekly

<https://dimario.dev>